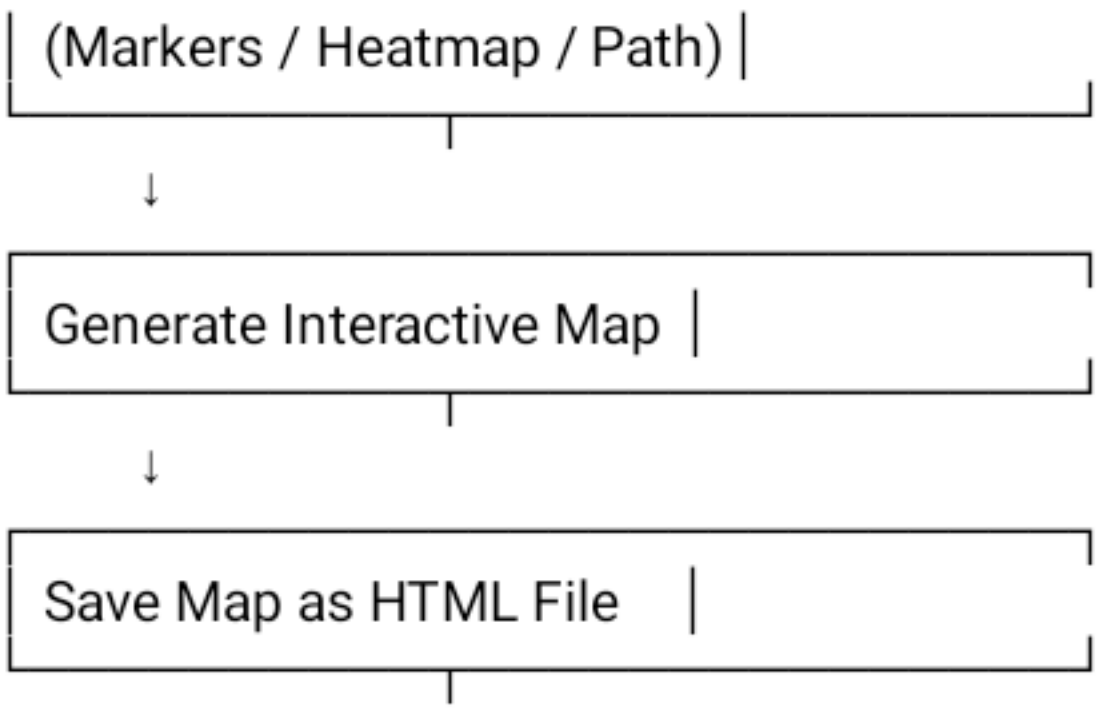
TITLE: 360° virtual map

Flow chart

```
┌──────────────────────────────────────┐
│ Select Image Folder                  │
└──────────────────────────────────────┘
                  │
                  ↓
┌──────────────────────────────────────┐
│        Extract EXIF Metadata         │
│          (Lat, Lon, Time)            │
└──────────────────────────────────────┘
                  │
                  ↓
┌──────────────────────────────────────┐
│     Convert GPS Coordinates          │
│        (DMS → Decimal)               │
└──────────────────────────────────────┘
                  │
                  ↓
┌──────────────────────────────────────┐
│        Handle Missing Data           │
│          (Interpolation)             │
└──────────────────────────────────────┘
                  │
                  ↓
┌──────────────────────────────────────┐
│        Store Data in Table           │
│          (DataFrame)                 │
└──────────────────────────────────────┘
                  │
                  ↓
┌──────────────────────────────────────┐
│ Create Base Map                      │
│ (Using Folium)                       │
└──────────────────────────────────────┘
                  │
                  ↓
┌──────────────────────────────────────┐
│ Add Map Layers                       │
```



SYNOPSIS

1.objective

The main objective of this project is to:

- Extract **location and time data from images**
- Convert GPS coordinates into usable format
- Generate an **interactive map** showing image locations
- Build a **CLI-based tool** for automation

2. Introduction

Digital images contain hidden information known as **EXIF (Exchangeable Image File Format)**, which stores metadata such as GPS location and timestamp.
This project uses Python libraries like exifread, pandas, GeoPandas, and Folium to extract, process, and visualize this data on an interactive map.
The system follows an **ETL (Extract, Transform, Load)** approach to convert unstructured image data into structured spatial information.

3. System Architecture

The project is divided into three main stages:

- **Extraction**
  - Read images from folders
  - Extract EXIF metadata (Latitude, Longitude, Time)
- **Transformation**
  - Convert GPS from **DMS to Decimal Degrees**
  - Handle missing data using interpolation
  - Store data in structured format
- **Visualization**
  - Create maps using Folium
  - Add markers, paths, and heatmaps

4. Methodology

Step 1: Image Collection

- Input folder containing images (.jpg, .png)

Step 2: Metadata Extraction

- Use EXIF library to extract GPS and timestamp

Step 3: Data Conversion

- Convert coordinates using:
  - Decimal = D + (M/60) + (S/3600)

Step 4: Data Cleaning

- Handle missing values using interpolation

Step 5: Data Storage

- Store in DataFrame for easy processing

Step 6: Map Creation

- Generate base map using Folium

Step 7: Visualization Layers

- Markers (locations)
- Heatmap (density)

- Path (movement tracking)

Step 8: Output

- Save map as HTML file

5. Tools & Technologies Used

- **Programming Language:** Python
- **Libraries:**
  - o exifread – Extract metadata
  - o pandas – Data handling
  - o GeoPandas – Spatial analysis
  - o Folium – Map visualization

6. Key Features

- Automatic extraction of image metadata
- Conversion of GPS coordinates
- Handling missing GPS data
- Interactive map generation
- Multiple visualization layers
- CLI-based execution

7. Applications

- **Infrastructure:** Damage tracking and inspection
- **Retail:** Store mapping and distribution planning
- **Environment:** Field data collection and research
- **Insurance:** Location verification of claims

8. Future Enhancements

- Support for more file formats (TIFF, RAW)
- Export to GeoJSON/KML
- Integration with web applications

- Real-time GPS tracking
- AI-based image classification

9. Conclusion

This project successfully converts image data into an interactive spatial map. It demonstrates how metadata can be used for real-world applications such as tracking, analysis, and decision-making.
The system is efficient, scalable, and useful across multiple industries.

10. Keywords

EXIF Metadata, GPS Coordinates, DMS to Decimal, Spatial Mapping, Interactive Maps, CLI Tool